Swiss Institute of Bioinformatics

First Steps with Python in Life Sciences

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General Information

Course page: https://github.com/sib-swiss/first-steps-with-Python-training

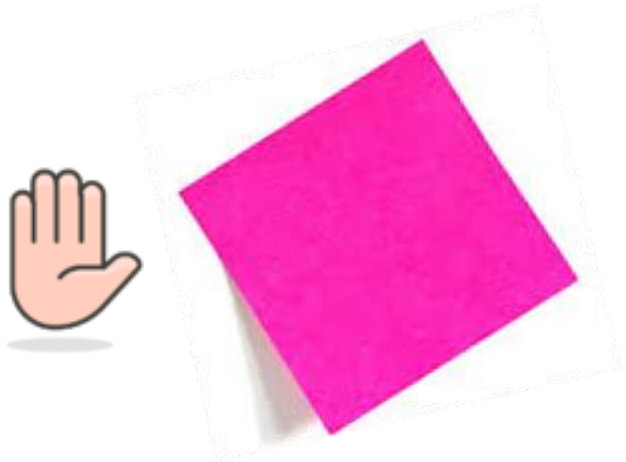
- Slides
- Data sets
- Notebooks
- Solutions

ELIXIR Code of Conduct

Optional exam, 0.5 ECTS value

Asking questions - Communication

- Raise your hand anytime



- Done with an exercise?

Course schedule

Day 1

- Morning: introduction to jupyter notebook, basic types
- Afternoon: container types

Day 2

- Morning: conditional statements (if…else), loops (for and while)
- Afternoon: functions, and reading/writing files

Day 3

- Morning : loading external modules
- Afternoon: additional modules and personal projects

Day 3 afternoon

Elective work on this last afternoon:

- Optional advanced modules
- Personal project work
- Consolidate python basics

We will detail the optional modules later, but you can already take a look.

Don't hesitate to prepare questions if you have a project in mind already.

Introducing Ourselves

Who are you?

Why do you want to program?

Have you programmed before? Which language?

What is Python?

"Python is an **interpreted**, **high-level**, **general-purpose** programming language."
wikipedia

Interpreted: no compilation of the program prior to execution

- + platform independence (portability), dynamic typing
- - usually slower, buggy program may still run

High level: abstracted from details of the machine (e.g. memory management)

- + focus on the application itself
- - possibly counter-intuitive behaviors (e.g. mutable vs. immutable objects).

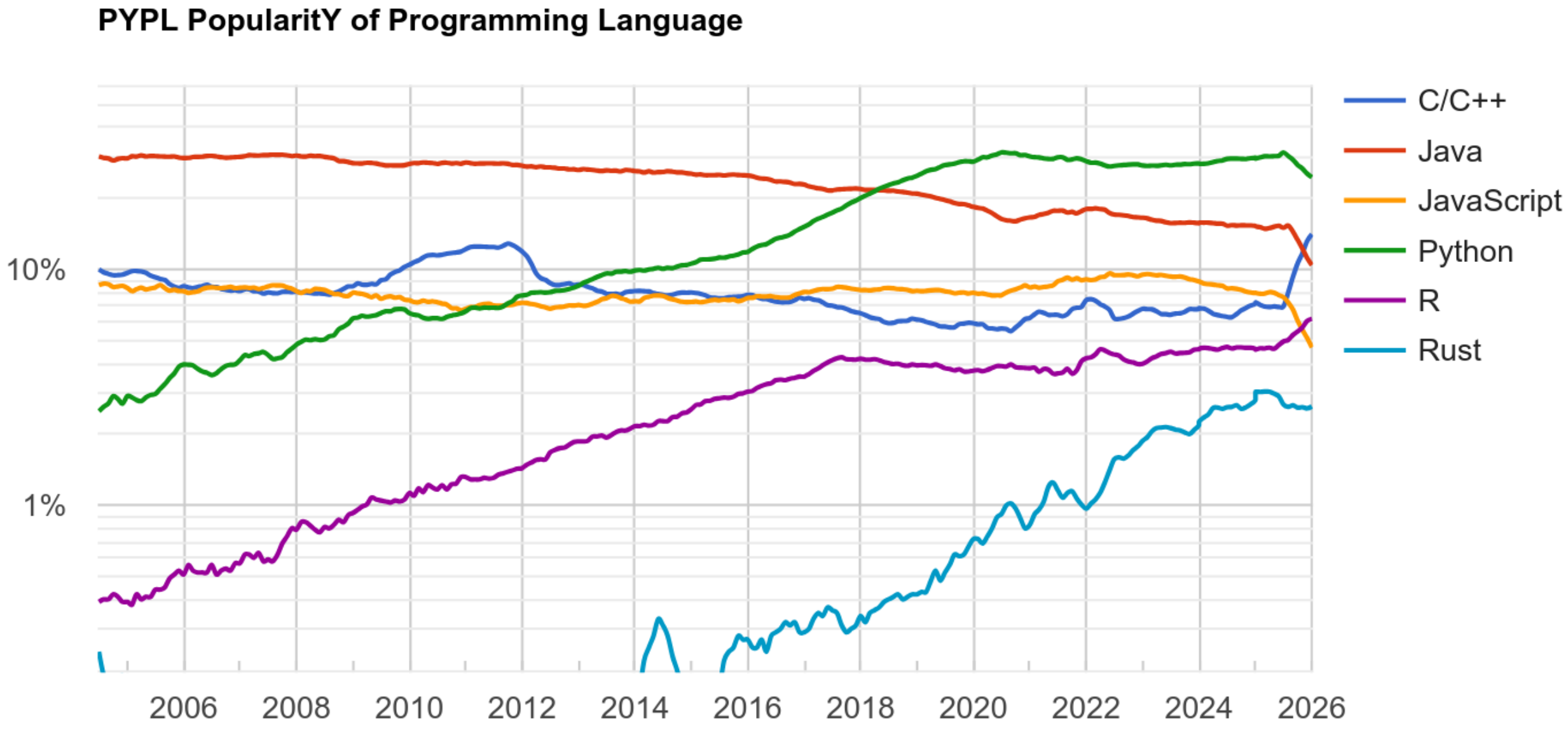
General-purpose: not domain specific, can be used in a broad range of software application

- + wide user base, usable for any purpose
- - core language fairly simple → modules for domain specific uses

Python – a brief history

- **1991**: First version of Python (0.9.0) publicly released by Guido van Rossum.
  - Its name is a tribute to the British comedy group “Monty Python”, it later adopted the two snakes as logo symbol.
- **2000**: Python 2.0
- **2008**: Python 3.0 (backward incompatible with python2)
- ~~**2015**~~ **2020** : end of Python 2.7 support
- **Current version** : 3.14 (as of January 2025)

Python – is it used?



https://pypl.github.io/PYPL.html

Python:
portable
broad range of applications
easy to learn
widely used

This course is the first step in your python journey

python™